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BIT 2223 Mobile Computing

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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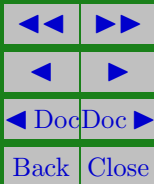
LESSON 2

Wireless Technologies

Objectives

By the end of this topic you should be able to;

- *Explain the major types of wireless networks WLAN, PAN, WMAN.*
- *Explain the concept of Blue-tooth technology.*
- *Understand the operational principles behind major wireless networks*
- *Understand the concept of spread spectrum*





2.1. Introduction

There are about eight types of network which are used world wide these days, both in houses and commercially. These networks are used on the bases of their scale and scope, historical reasons, preferences for networking industries, and their design and implementation issues. LAN and WAN are mostly known and used widely. LAN, local area network was first invented for communication between two computers. LAN operates through cables and network cards. Later WLAN, Wireless local area network was formed through LAN concept, there are no wires involved in communication between computers, and Wireless LAN cards are required to connect to wireless network. LAN is the original network out of which other networks are formed according to requirements. They are as follows:



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- LAN - Local Area Network
- WLAN - Wireless Local Area Network
- WAN - Wide Area Network
- MAN - Metropolitan Area Network
- SAN - Storage Area Network, It can also be referred with names like System Area Network, Server Area Network, or sometimes Small Area Network
- CAN - Campus Area Network, Controller Area Network, and often Cluster Area Network
- PAN - Personal Area Network
- DAN - Desk Area Network



2.2. Classification of wireless network

2.2.1. LAN - Local Area Network

LAN connects networking devices within a short span of area, i.e. small offices, home, internet cafes etc. LAN uses TCP/IP network protocol for communication between computers. It is often but not always implemented as a single IP subnet. Since LAN is operated in a short area, it can be controlled and administered by a single person or organization.

2.2.2. Wireless PAN

Wireless Personal Area Network (WPAN) is a type of wireless network that interconnects devices within a relatively small area, generally within reach of a person as used by the likes of infrared



and Blue-tooth technologies. For example, Blue-tooth provides a WPAN for interconnecting a headset to a laptop.

Personal Area Networks (PAN) are formed by wireless communications between devices by way of technologies such as Blue-tooth and UWB. PAN standards are embodied by the IEEE 802.15 family of "Home Wireless" standards, which supersedes older infrared standards and HomeRF for dominance in this area of wireless communications.

● Applications

PAN applications are virtually limitless. However, a key capability for a PAN is to enable devices to autonomously detect and acquire one another. This provides the ability for personalization through unconscious communications. For example,



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a PAN in ones automobile would provide various applications, such as the ability for the system to detect the presence of the user, thereby allowing the mobile handset to automatically acquire pertinent information for driving such as weather and road conditions.

One of the fundamental advantages of PANs is that they negate the need for wires, allowing the user to, for example, have a wireless headset, create ad hoc connections to other applications, include mobile commerce in which user of a mobile device communicates with another machine for commerce such as ticket purchase, vending and other small purchases.

The evolution of PANs will be from man-to-man, man-to-machine, and finally machine-to-machine.



- **Issues and Challenges with PANs**

The biggest initial issue will simply be to equip devices with software to enable the PAN connection. This will occur once technology such as Blue-tooth is cost effective and available in large quantities for deployment.

One of the biggest issues with PANs is the ability for devices to inter-operate with one another. This is not so much an issue with pre-established networks of devices, which all have the same vendor equipment, but it is a major issue for inter-vendor equipment connections. This is a major factor for unconscious communications. Companies are working to solve these issues with various equipment mediation and interoperability software.



- **Ad hoc vs. Unconscious Communications**

Ad hoc connections are simply those that occur in an automatic fashion, but the user is engaged in some activity in which it is known that a connection is about to occur. Examples include connecting to a LAN or buying a beverage from a vending machine.

On the other hand, unconscious communications occur as a result of automatic connections that are made between devices unknown to the user. Examples include a user coming home and the local base station in the home sensing this event and thus messaging to the network to treat all calls as home zone calls rather than mobile calls in the macro network. Unconscious connections are the real end-goal for many applications that involve personalization and mobile commerce. The real issue will



be to solve the interoperability problems between differing vendor provided equipment.

● PAN Standards

- 802.15.1a: Blue-tooth (2.4GHz at 1Mbps)
- 802.15.2: Coexistence of PANs with one another
- 802.15.3: High rate PAN (2.4GHz at 55 Mbps)
- 802.15.3a: Alternative high rate PAN for UWB (2.4GHz at 110 Mbps)
- 802.15.4: Low rate PAN - Zigbee
- 802.15.4a: Alternative low rate - low power UWB



2.2.3. MAN - Metropolitan Area Network

This kind of network is not mostly used but it has its own importance for some government bodies and organizations on larger scale. MAN, metropolitan area network falls in the middle of LAN and WAN. It covers a large span of physical area than LAN but smaller than WAN, such as a city.

2.2.4. CAN - Campus Area Network

This kind of network is mostly used in relatively large universities or local business offices and buildings. Its networking is spanned with multiple LANs but smaller than a Metropolitan area network, MAN.



2.2.5. SAN - Storage Area Network

SAN technology is used for data storage and it has no use for most of the organization but data oriented organizations. Storage area network connects servers to data storage devices by using Fiber channel technology.

- **SAN - System Area Network**

SAN, system area networks are also known as cluster area network and they connect high performance computers with high speed connections in cluster configuration.

2.2.6. Wireless LAN

Wireless Local Area Network (WLAN) is a wireless alternative to a computer Local Area Network (LAN) that uses radio in-



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stead of wires to transmit data back and forth between computers in a small area such as a home, office, or school.

Wireless Local Area Networks (WLAN) are increasingly becoming an attractive alternative to licensed spectrum. Initially thought of strictly as a competitor to 3G, WLAN is now thought of as a complement to cellular based data services, providing a high bandwidth alternative to 3G at a fraction of the cost. Wireless LANs are standardized under the IEEE 802.11 series.

- **Wi-Fi:** Wi-Fi is a commonly used wireless network in computer systems to enable connection to the internet or other devices that have Wi-Fi functionalities. Wi-Fi networks broadcast radio waves that can be picked up by Wi-Fi receivers attached to different computers or mobile phones.



- **Fixed Wireless Data:** This implements point to point links between computers or networks at two locations, often using dedicated microwave or laser beams over line of sight paths. It is often used in cities to connect networks in two or more buildings without physically wiring the buildings together.

Being that WLAN's are only available through limited access points referred to as hot spots, it is important to recognize that WLAN service does not allow for truly ubiquitous service. **Public Access WLAN** requires deployment of certain network capabilities including access control, authentication, authorization, and usage accounting. In addition, clearing and settlement among businesses in the value chain is required, as access point providers, serving networks, and home networks all need to be



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paid appropriately. Only through the implementation of these capabilities may a service provider effectively provide WLAN roaming to its customers.

One of the key aspects of providing public WLAN is to offer comprehensive billing services. Many services only offer the ability to charge to one's credit card. As services evolve, the need to offer billing to one's cellular service provider bill or other billing alternative will become increasingly important.

Makes use of wireless transmission medium. provide wireless network communication over short distances using radio or infrared signals instead of traditional network cabling. Until recently wireless LANS were of little use. The reason included high price, low data rates, occupation safety concerns and licensing requirement.



- **3G and WLAN**

One of the more important issues associated with WLAN roaming is the ability to allow roaming between 3G and WLAN services. This is a key capability to allow for a seamless service experience between all the (licensed and unlicensed spectrum based) services that a mobile network operator may offer.

WLAN roaming may utilize some of the same GRX-type infrastructure as used for GPRS or 1xRTT roaming, thus providing synergies and economies of scale and scope to common service providers.

- **Application Areas**

- **LAN Extension:** Wireless LAN- linked into a wired LAN on same premises; Wired LAN Backbone, Support servers



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and stationary workstations; Wireless LAN-Stations in large open areas, Manufacturing plants, stock exchange trading floors, and warehouses.

- **Cross-building interconnect:** Connect LANs in nearby buildings-a point to point wireless link is used between two buildings. Device connected are typically bridges and routers.
- **Nomadic Access:** Wireless link between LAN hub and mobile data terminal equipped with antenna; Laptop computer or notepad computer

Uses:

- Transfer data from portable computer to office server



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- Can be useful in extended environment such as campus
- Ad hoc networking
- **Ad Hoc Networking** - (no infrastructure): Temporary peer-to-peer network set up to meet immediate needs; Example: Group of employees with laptops convene for a meeting and later link computers in a temporary network for duration of meeting.

In summary, WLAN forms a stationary infrastructure consisting of one or more cells with a control module for each cell. Within a cell there could be a number of end stations. Nomadic stations can move from one cell to another. There is no infrastructure for adhoc networks. It is a peer to peer collection of stations within range of each other which may dynamically configure themselves



into a temporary network.

2.3. Wireless LAN Requirements

- *Throughput*: Medium access control protocols should make efficient use as possible of the medium access to maximize capacity.
- *Number of nodes*: Wireless nodes needs to support hundreds of nodes across multiple cells.
- *Connection to backbone LAN*: Interconnection with wired backbone should be possible.
- *Service area*: Typical coverage of wireless LANs has a diameter of 100 to 300m.
- *Battery power consumption*: long life battery.(MAC pro-



protocol require MS to monitor access point constantly or engage in handshake with BS). WLAN have features to reduce power consumption while not using the network, such as sleep mode.

- *Transmission robustness and security*: Tackle challenges of noise and other interference. Provide some level of security from eavesdropping.
- *License free operation*: users would prefer to buy and operate wireless LANs products without having to secure license for frequency band used by the LAN.
- *Hand-off/roaming*: MAC protocol used in wireless LAN should allow MS to move from cell to cell.
- *Dynamic configuration*: MAC addressing and network man-



agement aspect should permit dynamic and automatic addition, deletion and relocation of end station without disruption to other users.

2.4. Wireless LAN Categories

- **Infrared (IR) LANs:** An individual cell of an IR LAN is limited to single room because Infrared does not penetrate opaque walls.
- **Spread spectrum LANs:** This type of LAN makes use of spread spectrum transmission technology. These LAN operate in ISM band (Industrial science and medical) bands which are unlicensed.
- **Narrow-band microwave:** These LANS operate at microwave frequencies. Some operate at frequencies that re-



quire licensing while others use ISM bands.

2.4.1. WAN - Wide Area Network

As the “word” Wide implies, WAN, wide area network cover large distance for communication between computers. The Internet itself is the biggest example of Wide area network, WAN, which is covering the entire earth. WAN is a distributed collection of geographically LANs. A network connecting device router connects LANs to WANs. WAN uses network protocols like ATM, X.25, and Frame Relay for long distance connectivity.

2.4.2. Wireless MAN

Wireless Metropolitan Area Networks are a type of wireless network that connects several Wireless LANs. WiMAX (from



”Worldwide Interoperability for Microwave Access”) is the term which has been commercialized to refer to wireless MANs and is covered in IEEE 802.16d/802.16e.

2.4.3. PAN

A personal area network (PAN) is a computer network used for communication among computer devices (including telephones and personal digital assistants) close to one’s person. The devices may or may not belong to the person in question. The reach of a PAN is typically a few meters.

Personal area networks may be wired with computer buses such as USB and FireWire. A wireless personal area network (WPAN) can also be made possible with network technologies such as IrDA, Blue-tooth, UWB, Z-Wave and ZigBee



2.5. WiMAX

WiMAX is new technology in mobile computing. It is yet to be deployed on huge scale world wide unlike technologies like Wi-Fi. WiMAX is short form of Worldwide Interoperability of Microwave Access. It performs is about similar to some extent like Wi-Fi but at higher speed, at great distance and for greater number of users, WiMAX operates on 802.16 standards defined by IEEE, Institute of Electric and Electronic Engineering. IEEE 802.16 standard is known to be WMAN, Wireless Metropolitan Area Network.

Using WiMAX technology, broadband access can be provided to more rural and urban areas where wireless services cannot be provided by other sources like fiber optics or by other regular telephonic cables. There are so many plus points of this



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technology out of which one is range, its range starts from minimum 30 miles, about 30 feet wireless connectivity is provided by Wi-Fi technology. WiMAX range depends upon connectivity, if the connection is with Line of Sight (LoS) it can increase from up to 75 miles. WiMAX offers a solution called “Last mile” it can be used as alternative of cable and DSL internet access.

WiMAX can provide data rate up to 70Mbps from larger distances which be can be reached up 30 miles. WiMAX operates on two frequencies.

1. IEEE 802 is a very flexible standard and provides standard addresses frequencies ranges from 10 GHz to 66 GHz.
2. IEEE 802.16 standard frequency states that it will also support low latency applications like internet, video, voice all together.



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WiMAX technology is quite new, therefore few vendors are trying to develop infrastructure to develop the technology. Vendors like Intel and Fujitsu are leading providers for WiMAX chipset, called SoC. Customer premise equipment (CPE) are manufactured by using these WiMAX chips SoCs, this enables the equipment to access WiMAX base station or Wireless Broadband Access (WBA).

2.5.1. How WiMAX Works

WiMAX technology works a little different than Wi-Fi technology. In Wi-Fi, computers can be connected through wireless LAN card to a nearby access point, wireless router or any Hotspot, it does not works this way when we talk about connectivity in WiMAX. WiMAX network connectivity constitutes of



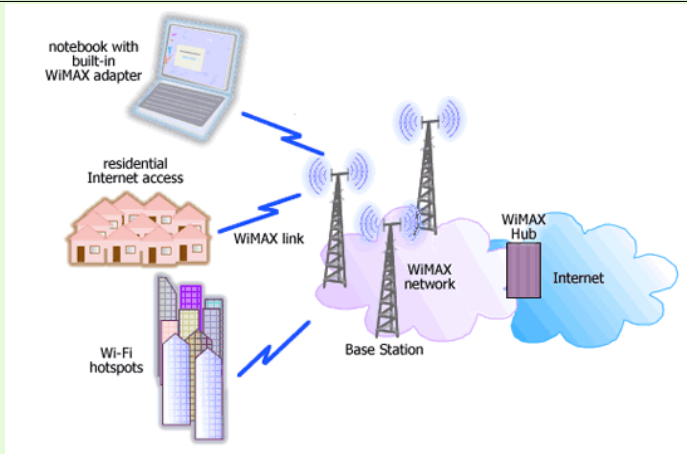
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two parts, one can define as WiMAX tower or WiMAX booster, it is known as WiMAX Base station, whereas the other portion is WiMAX receiver.

1. **WiMAX base station:** As the name explains, base station is a place where WiMAX signals are broadcasted. It consists of electronic devices and WiMAX Tower. This tower works exactly like GSM network phones towers standing high up in the air to broadcast radio signals. WiMAX tower base station can cover up 10Km radius. In theory it suggests to cover a lot more distance than just 10Km, it can reach some where about 50 km (30 miles), but in fact due to certain geographical limitations it goes as far as 10 km approx. 6 miles. Any wireless connecting device for WiMAX will connect to WiMAX network if it falls in the range.



2. **WiMAX Receiver:** It is a device or devices which receives the signals from WiMAX base station and connects to the WiMAX networks. These devices are usually stand alone, Antenna or PCMCIA slot card for laptops or computers. Connecting to WiMAX base stations works as similar as connection of Wi-Fi to access point works, the only difference is that WiMAX covers much wider area.



One WiMAX base stations can be connected to several other base stations using high speed microwave link, this link is usually known as **back haul**. This way WiMAX roaming can be achieved and connections can be maintained on move. WiMAX



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support many protocols like ATM, IPv4 Ethernet, VLAN etc, this makes WiMAX a rich choice for full of services from data to voice.

2.5.2. WiMAX Features

- **Long Range:** Connectivity from long range is a much highlighted feature of WiMAX.
- **Mobility:** Using its long range feature, it provides connectivity on areas where other means of high speed Internet are not available. The other major mobility feature of WiMAX technology is that connecting devices (Laptops, PDA etc) can move in certain area and still be connected to the network.
- **Interfacing:** WiMAX radio signals are broadcasted to its



subscribers from base stations.

- **Accessibility:** Accessing WiMAX base station is not a difficult task, its long range capability provides connectivity at home from base stations within the range. To get high speed internet, you only require to subscribe to WiMAX service providers.

2.6. BLUE-TOOTH

Blue-tooth is a standard and communications protocol primarily designed for: low power consumption, with a short range (1 meter, 10 meters, 100 meters) based on low-cost transceiver microchips in each device. It is designed to operate in an environment of many users. Up to 8 devices can communicate in a small area network called *piconet*. Ten piconets can co-exist in



the same coverage range of the Blue-tooth radio. For security, each link is encoded against eavesdropping and interference.

2.6.1. Blue-tooth Application Areas

- **Data and voice:** Access points facilitates real-time voice and data transmissions by providing effortless wireless connection of portable and stationary communication devices.
- **Cable replacement:** Blue-tooth eliminates need for numerous, often proprietary ,cable attachments connection for any kind of communication device.
- **Ad hoc networking:** Devices with Blue-tooth radio can establish connection with another when in range



- **Other prevalent applications of Blue-tooth.**

- Wireless control of and communication between a mobile phone and a hands-free headset or car kit. This was one of the earliest applications to become popular.
- Wireless networking between PCs in a confined space and where little bandwidth is required.
- Wireless communications with PC input and output devices, the most common being the mouse, keyboard and printer.
- Transfer of files between devices with OBEX. (object exchange protocol)
- Transfer of contact details, calendar appointments, and reminders between devices with OBEX.



2.6.2. Blue-tooth vs. Wi-Fi in networking

- Blue-tooth and Wi-Fi. Both use the unlicensed ISM (industrial, scientific and medical) band and apply spread spectrum technology.
- Wi-Fi provides higher throughput and covers greater distances, but requires more expensive hardware and higher power consumption.
- They use the same frequency range, but employ different multiplexing schemes.
- While Blue-tooth is a cable replacement for a variety of applications, Wi-Fi is a cable replacement only for local area network access.



2.6.3. ISM band

The industrial, scientific and medical (ISM) radio bands, defined by the ITU-R regulations, were originally reserved internationally for the use of RF electromagnetic fields for industrial, scientific and medical purposes other than communications. In general, communications equipment, using this band must accept any interference generated by ISM equipment

2.6.4. Blue-tooth Standards Documents

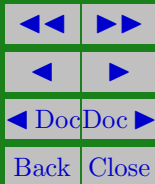
- **Core specifications:** Details of various layers of Blue-tooth protocol architecture
- **Profile specifications:** Use of Blue-tooth technology to support various applications



2.7. MAN

Although the 802.16 family of standards is officially called WirelessMan in IEEE, it has been commercialized under the name “WiMAX” (from “Worldwide Interoperability for Microwave Access”) by the industry alliance called the WiMAX Forum. The mission of the Forum is to promote and certify compatibility and interoperability of broadband wireless products based on the IEEE 802.16 standards.

The most popular implementation of the IEEE 802.16 standard is the Mobile WirelessMAN originally defined by the 802.16e-2005 amendment that is now in process of being deployed around the world in more than 140 countries by more than 475 operators





2.8. INFRARED LANS

Infrared LANS are strengths of waves over Microwave Radio. The Spectrum for infrared is virtually unlimited. Its equipment is inexpensive and simple. Infrared share some characteristics of visible light that makes it attractive for certain LAN configurations. i.e Infrared is diffusely reflected by light-colored objects thus it is possible to use ceiling reflection to achieve coverage for entire room coverage. Infrared does not penetrate walls. It is more easily secured against eavesdropping and has less interference between different rooms

2.8.1. Drawbacks of Infrared Medium

- Indoor environments experience infrared background radiation from sunlight and indoor lighting



- Ambient radiation appears as noise in an infrared receiver, requiring transmitters of higher power.
- Increased transmitter power is limited by concerns of eye safety and excessive power consumption
- LOS limitation (Limited range)

2.8.2. IR Data Transmission Techniques

1. Directed Beam Infrared

- Used to create point-to-point links. In this mode, the range depends on emitted power and degree of focusing
- Focused IR data link can have range of kilometers and can be used for cross building interconnect between bridges or routers



2. Omni-directional

- Involves single base station within line of sight of all other stations on LAN
- Base station typically mounted on ceiling and the station acts as a multi-port repeater
 - Ceiling transmitter broadcasts signal received by IR transceivers
 - IR transceivers transmit with directional beam aimed at ceiling base unit

3. Diffused

- All IR transmitters are focused and aimed at a point on diffusely reflecting ceiling. IR radiation striking the ceiling is re-radiated omni directionally and picked up by all receivers in the area.



2.9. Spread Spectrum

All wireless communications depends on sending and receiving signals broadcast through the atmosphere to carry information between devices. The signals take the form of electromagnetic spectrum. The frequency of wave form used for communication is hertz (hz) in honor of Heinrich Hertz., one of the inventors of radio. The spectrum starts with very low frequency for instance used by electrical power (60hz) and telephone 0-3khz (tradition voice system). Frequency affects the amount and speed of data transmission. The transmission strength or power determines the distance that broadcast data can travel and still remain intelligible. The principle governing wireless transmission still dictates that lower frequency transmission can carry less data more slowly over longer distances and high frequency can



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carry data faster over short distances. The middle part of electromagnetic spectrum is commonly divided into several named frequency ranges or bands.

The following names are commonly used to identify the portions:

- 3 kHz to 30 kHz Very Low Frequencies (VLF)
- 30 kHz to 300 kHz Low Frequencies (LF)
- 300 kHz to 3,000 kHz Medium Frequencies (MF)
- 3,000 kHz to 30,000 kHz High Frequencies (HF)
- 30,000 kHz to 300,000 kHz Very High Frequencies (VHF)
- 300,000 kHz to 3,000,000 kHz Ultra High Frequencies (UHF)

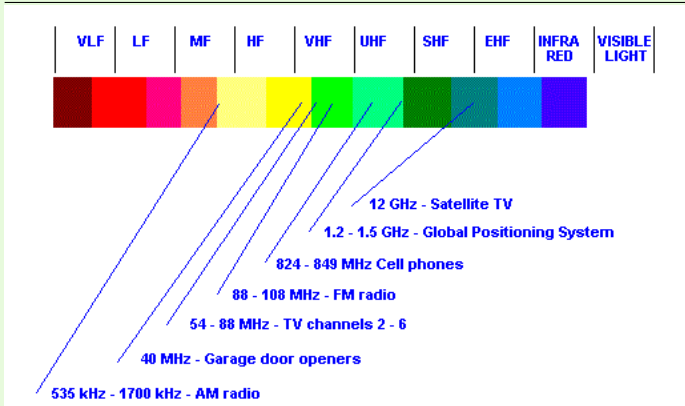


Fig2: Radio Frequencies

Common radio frequency bands include the following:

- AM radio - 535 kilohertz to 1.7 megahertz
- Short wave radio - bands from 5.9 megahertz to 26.1 megahertz



- Citizens band (CB) radio - 26.96 megahertz to 27.41 megahertz
- Television stations - 54 to 88 megahertz for channels 2 through 6
- FM radio - 88 megahertz to 108 megahertz
- Television stations - 174 to 220 megahertz for channels 7 through 13

2.9.1. General Frequency Ranges

1. Microwave frequency range

- 1 GHz to 40 GHz
- Directional beams possible
- Suitable for point-to-point transmission



- Used for satellite communications
- 2. Radio frequency range
 - 30 MHz to 1 GHz
 - Suitable for omni directional applications
- 3. Infrared frequency range
 - Roughly, 3×10^{11} to 2×10^{14} Hz
 - Useful in local point-to-point multi-point applications within confined areas

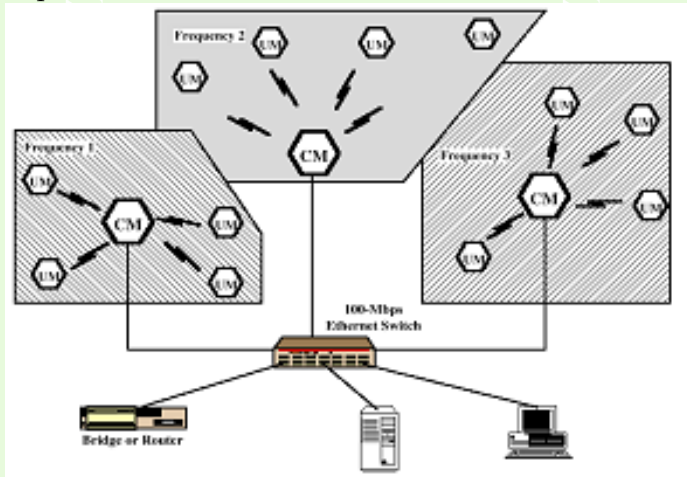
2.9.2. Spread Spectrum LAN Configuration

Spread spectrum uses Multiple-cell arrangement. Adjacent cells make use of different centre frequencies within some bands to avoid interferences. Within a cell, topologies can be either peer-to-peer or hub (Peer-to-peer topology does not have a hub). Its



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access is controlled with a MAC algorithm - CSMA which is appropriate for ad hoc LANs.



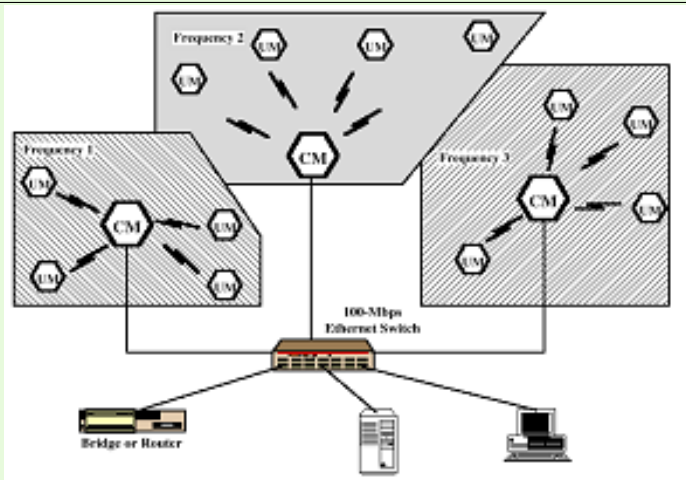


Fig3. Multiple Cell Wireless LAN configuration



2.9.3. Narrowband Microwave LANs

It refers to use of a microwave radio frequency band for signal transmission relatively narrow bandwidth both Licensed and Unlicensed. It can be licensed within specific geographic areas to avoid potential interference e.g Motorola - 600 licenses in 18-GHz range which covers all metropolitan areas. It assures that independent LANs in nearby locations don't interfere.



Revision Questions

EXERCISE 1.  Describe radio spectrum (radio frequency) by giving examples of applications in various bands.

Example . Give the comparison between Ad hoc and infrastructure

Solution: Infrastructure mode provides much more stability, scalability, ease of management and improved security. Ad hoc on the other hand does not provide security to that level and managing can be difficult incase of network growth. Performance suffers as devices are increased. □

EXERCISE 2.  Compare WiMAX Versus Wi-Fi



Solutions to Exercises

Exercise 1.

The radio spectrum is a subset of the electromagnetic waves lying between the frequencies from 9 kilohertz (kHz - thousands of cycles per second) to 30 gigahertz (GHz - billions of cycles per second). These support a wide range of business, personal, industrial, scientific, medical research and cultural activities, both public and private. It is that part of the natural spectrum of electromagnetic radiation lying between the frequency limits of 9 kilohertz and 300 gigahertz.

Exercise 1



Exercise 2.

Wi-Fi: Wi-Fi is used for developing wireless LAN to access high speed internet or access just a network for file sharing and software services.

WiMAX: WiMAX is a latest technology and it can do far more than just developing wireless networks for high speed internet. It is also refer as Wireless broadband access, which can transfer not only data, but voice data, video data etc and at much higher rates..

Wi-Fi: Wi-Fi is capable for short range data transfer, which can be within hundred of meters range using non licensed spectrum to access network. Wi-Fi is mostly connected to network in certain area which may not be connected to internet; it can



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be used for file sharing only.

WiMAX: WiMAX is designed for long distance coverage and it covers distance in kilometers, it uses licensed spectrum and unlicensed also in some case. WiMAX delivers point-to-point connection to the internet from service provide to as user. In WiMAX there are multiple standards of 802.16 a, b etc, so they are used for different types of access from mobile connectivity to fixed location connections.

Wi-Fi: Wi-Fi has introduced a quality of service similar to fixed Ethernet, where packets are priorities on their tags. This shows that quality of service (QoS) is relative to packet flow.

WiMAX: WiMAX uses technology based on setting up connection between end users' device and base station. Special algorithm is scheduled for specific connection. This shows that



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Quality of Services (QoS) parameters can be guaranteed for each flow

Wi-Fi is deployed much more than WiMAX because of its ease of installation and cost effectiveness. It is easily deployed within the building or room for providing internet access by third party internet service providers. Many public places, hotels, coffee shops have installed Wi-Fi access points providing high speed internet to their customers

Exercise 2